VANISHING OF COHOMOLOGY FROM THE VANISHING OF CECH COHOMOLOGY OVER A COFINAL COLLECTION OF COVERINGS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument for the criterion; the reduction to the Cech computation and the induction step are both inside the counted span.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Cech cohomology of a covering of a site. Same chapter, the Cech complex of a covering and its cohomology. Supplies the hypothesis of the statement and the complexes used in the proof.
- Comparison between Cech cohomology and cohomology for a cofinal system of coverings. Same chapter, the result that cohomology is computed by the Cech cohomology of a cofinal system of coverings. Supplies the first step, in which the cohomology of F is written as the colimit of the Cech cohomologies over the coverings in Cov.
- Long exact cohomology sequence of a short exact sequence of abelian sheaves. Same chapter, the long exact sequence attached to a short exact sequence of abelian sheaves. Supplies the induction step, in which the vanishing in degree p is deduced from a suitable short exact sequence and the hypothesis on Cech cohomology.

Source reference. Original lemma 01EV: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 1566–1566. Navigation pointer invoked at proof line 1566; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Date: stacks-project:sites-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 00VH: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 1569–1569. Navigation pointer invoked at proof line 1569; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 03AW: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 1334–1334. Navigation pointer invoked at proof line 1593; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 03F8: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 1524–1524. Navigation pointer invoked at proof line 1600; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 015B: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 1639–1639. Navigation pointer invoked at proof line 1639; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *(Variant of Cohomology, Lemma 01EV.) Let $\mathcal{C}$ be a site. Let $\mathrm{Cov}_{\mathcal{C}}$ be the set of coverings of $\mathcal{C}$ (see Sites, Definition 00VH). Let $\mathcal{B} \subset \mathrm{Ob}(\mathcal{C})$, and $\mathrm{Cov} \subset \mathrm{Cov}_{\mathcal{C}}$ be subsets. Let $\mathcal{F}$ be an abelian sheaf on $\mathcal{C}$. Assume that*

1. *For every $\mathcal{U} \in \mathrm{Cov}$, $\mathcal{U} = \{U_i \to U\}_{i \in I}$ we have $U, U_i \in \mathcal{B}$ and every $U_{i_0} \times_U \ldots \times_U U_{i_p} \in \mathcal{B}$.*
2. *For every $U \in \mathcal{B}$ the coverings of U occurring in Cov is a cofinal system of coverings of U.*
3. *For every $\mathcal{U} \in \mathrm{Cov}$ we have $\check{H}^p(\mathcal{U}, \mathcal{F}) = 0$ for all $p > 0$.*

Then $H^p(U, \mathcal{F}) = 0$ for all $p > 0$ and any $U \in \mathcal{B}$.

Proof. Let $\mathcal{F}$ and Cov be as in the lemma. We will indicate this by saying "$\mathcal{F}$ has vanishing higher Čech cohomology for any $\mathcal{U} \in \mathrm{Cov}$". Choose an embedding $\mathcal{F} \to \mathcal{I}$ into an injective abelian sheaf. By Lemma 03AW $\mathcal{I}$ has vanishing higher Čech cohomology for any $\mathcal{U} \in \mathrm{Cov}$. Let $\mathcal{Q} = \mathcal{I}/\mathcal{F}$ so that we have a short exact sequence

$$0 \to \mathcal{F} \to \mathcal{I} \to \mathcal{Q} \to 0.$$

By Lemma 03F8 and our assumption (2) this sequence gives rise to an exact sequence

$$0 \to \mathcal{F}(U) \to \mathcal{I}(U) \to \mathcal{Q}(U) \to 0.$$

for every $U \in \mathcal{B}$. Hence for any $\mathcal{U} \in \mathrm{Cov}$ we get a short exact sequence of Čech complexes

$$0 \to \check{\mathcal{C}}^\bullet(\mathcal{U}, \mathcal{F}) \to \check{\mathcal{C}}^\bullet(\mathcal{U}, \mathcal{I}) \to \check{\mathcal{C}}^\bullet(\mathcal{U}, \mathcal{Q}) \to 0$$

since each term in the Čech complex is made up out of a product of values over elements of $\mathcal{B}$ by assumption (1). In particular we have a long exact sequence of Čech cohomology groups for any covering $\mathcal{U} \in \mathrm{Cov}$. This implies that $\mathcal{Q}$ is also an abelian sheaf with vanishing higher Čech cohomology for all $\mathcal{U} \in \mathrm{Cov}$.

Next, we look at the long exact cohomology sequence

$$\begin{array}{ccccccc} 0 & \longrightarrow & H^0(U,\mathcal{F}) & \longrightarrow & H^0(U,\mathcal{I}) & \longrightarrow & H^0(U,\mathcal{Q}) \\ & & & \searrow & & & \\ & & H^1(U,\mathcal{F}) & \longrightarrow & H^1(U,\mathcal{I}) & \longrightarrow & H^1(U,\mathcal{Q}) \\ & & & \searrow & & & \\ & & \ldots & & \ldots & & \ldots \end{array}$$

for any $U \in \mathcal{B}$. Since $\mathcal{I}$ is injective we have $H^n(U,\mathcal{I}) = 0$ for $n > 0$ (see Derived Categories, Lemma 015B). By the above we see that $H^0(U,\mathcal{I}) \to H^0(U,\mathcal{Q})$ is surjective and hence $H^1(U,\mathcal{F}) = 0$. Since $\mathcal{F}$ was an arbitrary abelian sheaf with vanishing higher Čech cohomology for all $\mathcal{U} \in \mathrm{Cov}$ we conclude that also $H^1(U,\mathcal{Q}) = 0$ since $\mathcal{Q}$ is another of these sheaves (see above). By the long exact sequence this in turn implies that $H^2(U,\mathcal{F}) = 0$. And so on and so forth. $\square$